Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Ratios and proportions

Simplify the ratio $18 : 24$ to lowest terms.

Divide both terms by 6: $3 : 4$.

2 [Foundational] • Target: Ratios and proportions

Solve the proportion $\frac{5}{8} = \frac{x}{40}$.

Cross-multiply: $8x = 200$, so $x = 25$.

3 [Foundational] • Target: Ratios and proportions

A recipe requires flour and sugar in the ratio $3 : 2$. If 12 cups of flour are used, how much sugar is needed?

Sugar $= \frac{2}{3} \times 12 = 8$ cups.

4 [Foundational] • Target: Unit rates

A car uses 8 gallons of gas to travel 224 miles. Find its fuel efficiency in miles per gallon.

$\frac{224}{8} = 28$ miles per gallon.

5 [Foundational] • Target: Unit rates

A printer produces 150 pages in 6 minutes. Find its rate in pages per minute, and use it to find how long it takes to print 400 pages.

Rate $= \frac{150}{6} = 25$ pages per minute. Time $= \frac{400}{25} = 16$ minutes.

6 [Foundational] • Target: Percentage change and applications

A price increases from 60 to 75 dollars. Find the percentage increase.

$\frac{75 - 60}{60} \times 100 = 25\%$.

7 [Foundational] • Target: Percentage change and applications

A jacket originally 120 dollars is discounted 30 percent. Find the sale price.

$120 \times 0.7 = 84$ dollars.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Percentage change and applications

A population decreases by 15 percent one year, then increases by 15 percent the next. Is the final population the same as the original?

Start with 100. After a 15\% decrease: 85. After a 15\% increase: $85 \times 1.15 = 97.75$. No — the final population is 2.25\% less than the original.

9 [Intermediate] • Target: Percentage change and applications

An investment of 2000 dollars grows to 2300 dollars after one year. Find the percentage increase.

$\frac{2300 - 2000}{2000} \times 100 = 15\%$.

10 [Intermediate] • Target: Weighted averages

A student scores 80 on a test worth 40 percent of the grade and 90 on a test worth 60 percent of the grade. Find the weighted average grade.

$0.4(80) + 0.6(90) = 32 + 54 = 86$.

11 [Intermediate] • Target: Weighted averages

A store sells 3 items at 10 dollars each and 7 items at 20 dollars each. Find the average price per item sold.

Total revenue $= 3(10) + 7(20) = 30 + 140 = 170$ dollars. Total items $= 10$. Average $= \frac{170}{10} = 17$ dollars.

12 [Intermediate] • Target: Scale and similar figures

A model car is built at a scale of $1 : 24$. If the actual car is 4.8 meters (480 cm) long, find the length of the model, in centimeters.

$\frac{480}{24} = 20$ centimeters.

13 [Intermediate] • Target: Scale and similar figures

A map has a scale of 1 cm to 5 km. If two cities are 8.5 cm apart on the map, find the actual distance between them.

$8.5 \times 5 = 42.5$ km.

14 [Intermediate] • Target: Interpreting data tables

A table shows the hours worked by 4 employees in a week: A: 32, B: 40, C: 28, D: 36. Find the total hours worked and the mean hours per employee.

Total $= 32 + 40 + 28 + 36 = 136$. Mean $= \frac{136}{4} = 34$ hours.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Interpreting data tables** • **Cross-Lesson: 5.01 Statistics & Probability**

A table shows quarterly revenue, in thousands of dollars: Q1: 50, Q2: 65, Q3: 70, Q4: 55. Find the percentage of annual revenue earned in Q3, to the nearest tenth of a percent.

Total $= 50 + 65 + 70 + 55 = 240$. Q3 percentage $= \frac{70}{240} \times 100 \approx 29.2\%$.

16 **[Challenging]** • **Target: Bar graphs and data displays** • **Cross-Lesson: 6.01 Geometry**

The bar graph shows a company's monthly sales, in thousands of dollars. Find the percentage increase in sales from January to March.

$\frac{65 - 40}{40} \times 100 = 62.5\%$.

17 **[Challenging]** • **Target: Proportional reasoning in context** • **Cross-Lesson: 7.01 Trigonometry & Additional Topics**

A factory produces 450 units using 6 machines running for 8 hours. Assuming the same rate per machine-hour, how many units would 9 machines produce in 8 hours?

Rate $= \frac{450}{6 \times 8} = \frac{450}{48} = 9.375$ units per machine-hour. New machine-hours $= 9 \times 8 = 72$. Units $= 72 \times 9.375 = 675$.

18 **[Challenging]** • **Target: Proportional reasoning in context** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

If 5 workers can build a fence in 12 days, how many days would it take 4 workers, assuming the same work rate per worker?

Total work $= 5 \times 12 = 60$ worker-days. Days for 4 workers $= \frac{60}{4} = 15$ days.

19 **[Challenging]** • **Target: A well-designed word problem: customer growth for a small business** • **Cross-Lesson: 1.01 Heart of Algebra**

This question has three parts. In the first quarter, a business had 240 customers, and 180 of them made a repeat purchase. (a) Find the repeat-purchase rate as a percentage. (b) In the second quarter, the number of customers grew by 25 percent, and the repeat-purchase rate held steady at the percentage found in part (a). Find the number of repeat customers in the second quarter. (c) The business sets a goal to raise the repeat-purchase rate to 80 percent by the third quarter, while keeping the same total number of customers as the second quarter. Find how many additional repeat customers (beyond part (b)'s total) are needed to reach this goal.

(a) $\frac{180}{240} \times 100 = 75\%$. (b) New customers $= 240 \times 1.25 = 300$. Repeat customers $= 0.75 \times 300 = 225$. (c) 80\% of 300 is $0.8 \times 300 = 240$ repeat customers needed. Additional customers needed $= 240 - 225 = 15$.

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20 [Challenging] • Target: Unit rates • Cross-Lesson: 2.01 Quadratics & Polynomials

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

Step 1: Set up the weighted average inequality with final exam score F: $0.20(92) + 0.35(76) + 0.45(F) \geq 84$. Step 2: Compute the accumulated points: $0.20(92) = 18.4, \quad 0.35(76) = 26.6 \Rightarrow 18.4 + 26.6 = 45.0.$ Step 3: Isolate F: $45.0 + 0.45F \geq 84 \Rightarrow 0.45F \geq 84 - 45 = 39 \Rightarrow F \geq \frac{39}{0.45} = \frac{3900}{45} = \frac{260}{3} \approx 86.67.$ The student must score at least **86.7** (or $\frac{260}{3}$) on the Final Exam.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Ratios and proportions

A budget decreases from 40 to 20. Find the percentage decrease.

a. 50% ✓ **b.** 55%

c. 100% **d.** 45%

$\frac{20}{40} \times 100 = 50\%.$

22 [Foundational] • Target: Unit rates

A population increases from 220 to 264. Find the percentage increase.

a. 40% **b.** 20% ✓

c. 25% **d.** 15%

$\frac{44}{220} \times 100 = 20\%.$

23 [Foundational] • Target: Percentage change and applications

An attendance figure increases from 240 to 312. Find the percentage increase.

a. 30% ✓ **b.** 25%

c. 35% **d.** 60%

$\frac{72}{240} \times 100 = 30\%.$

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24 [Foundational] • Target: Weighted averages

A subscription count increases from 20 to 29. Find the percentage increase.

a. 40%

b. 45% ✓

c. 90%

d. 50%

$\frac{9}{20} \times 100 = 45\%$.

25 [Foundational] • Target: Scale and similar figures

A membership total increases from 60 to 78. Find the percentage increase.

a. 30% ✓

b. 35%

c. 25%

d. 60%

$\frac{18}{60} \times 100 = 30\%$.

26 [Foundational] • Target: Interpreting data tables

A membership total decreases from 60 to 30. Find the percentage decrease.

a. 55%

b. 100%

c. 45%

d. 50% ✓

$\frac{30}{60} \times 100 = 50\%$.

27 [Foundational] • Target: Bar graphs and data displays

A population decreases from 260 to 169. Find the percentage decrease.

a. 30%

b. 70%

c. 35% ✓

d. 40%

$\frac{91}{260} \times 100 = 35\%$.

28 [Foundational] • Target: Proportional reasoning in context

A price increases from 260 to 377. Find the percentage increase.

a. 40%

b. 90%

c. 45% ✓

d. 50%

$\frac{117}{260} \times 100 = 45\%$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: customer growth for a small business**

An attendance figure decreases from 300 to 180. Find the percentage decrease.

a. 80%

b. 45%

c. 35%

d. 40% ✓

$\frac{120}{300} \times 100 = 40\%$.

30 **[Intermediate]** • **Target: Ratios and proportions**

A student scores 60 on an assignment worth 75% of the grade and 100 on an assignment worth 25% of the grade. Find the weighted average grade.

a. 80

b. 70 ✓

c. 67

d. 73

$0.75(60) + 0.25(100) = 45 + 25 = 70$.

31 **[Intermediate]** • **Target: Unit rates**

If 9 workers can complete a task in 24 days, how many days would it take 8 workers, assuming the same work rate per worker?

a. 29

b. 27 ✓

c. 24

d. 25

Total work $= 9 \times 24 = 216$ worker-days. Days for 8 workers $= \frac{216}{8} = 27$.

32 **[Intermediate]** • **Target: Percentage change and applications**

The bar chart shows a small business's quarterly income (in 1000s) for one year. In which quarter did they earn the smallest income?

a. 1st quarter

b. 2nd quarter

c. **3rd quarter ✓**

d. 4th quarter

Q3 has the smallest bar at 35 (thousand dollars).

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33 [Intermediate] • Target: Weighted averages

The population of a town increased by 20% in the first year and then decreased by 15% in the second year. What was the net percentage change over the two years?

a. Increase of 2% ✓

b. Increase of 5%

c. Decrease of 2%

d. No change

Solving through rates and algebraic proportions yields Increase of 2%.

34 [Intermediate] • Target: Scale and similar figures

The price of gasoline drops by 15% and then rises by 20%. What is the net overall percentage change?

a. Increase of 2% ✓

b. Increase of 5%

c. Decrease of 2%

d. Increase of 1%

Solving through rates and algebraic proportions yields Increase of 2%.

35 [Intermediate] • Target: Interpreting data tables

A survey found that 34% of respondents trust TV News most, and 22% trust Radio most. If a similar survey was conducted on 120 people with the same percentage breakdown, approximately how many people would say radio or TV news is most trustworthy?

a. 54

b. 61

c. 67 ✓

d. 72

$(34\% + 22\%) \times 120 = 56\% \times 120 = 67.2 \approx 67.$

36 [Intermediate] • Target: Bar graphs and data displays

The bar chart shows the percentage of primary students who like each subject. Which subject had the lowest percentage of likers?

a. Math

b. English ✓

c. Science

d. History

English has the lowest bar at 41%.

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37 [Intermediate] • Target: Proportional reasoning in context

By what percentage should we raise the price of a good to return to its initial price knowing that it was subject to a 20% discount 2 weeks ago?

a. 15%

b. 20%

c. **25%** ✓

d. 30%

After a 20% discount, price is $0.8P$. To return to P: $P/0.8P = 1.25$, a 25% increase.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: customer growth for a small business • Cross-Lesson: 8.01 Exponentials & Logarithms

The bar chart below shows the number of students who participated in different sports. The number of students who participated in swimming is unknown and represented by x. If the total number of students who participated in all sports is 120, and the number of students who participated in basketball, volleyball, and soccer are 40, 30, and 20 respectively, what is the value of x?

a. 20

b. 30

c. **40** ✓

d. 50

The bar chart represents the number of students in different sports. Let the number of students in swimming be x. The number of students who participated in basketball is 40. The number of students who participated in volleyball is 30. The number of students who participated in soccer is 20. The total number of students who participated in all sports is 120. To find the value of x, we can set up an equation where the sum of students in each sport equals the total number of students: $x + 40 + 30 + 20 = 120$ Com.

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39 [Challenging] • Target: Ratios and proportions • Cross-Lesson: 1.01 Heart of Algebra

The bar chart below shows the number of students who participated in different sports at a school. What is the ratio of the number of students who play basketball to the number of students who play volleyball?

a. 1:2

b. 2:3

c. **3:2** ✓

d. 3:4

From the bar chart, we can see that 18 students play basketball and 12 students play volleyball. The ratio of the number of students who play basketball to the number of students who play volleyball is $18:12$. To simplify this ratio, we find the greatest common divisor of 18 and 12, which is 6. Divide both numbers by 6: $\frac{18}{6} = 3$ and $\frac{12}{6} = 2$. Therefore, the simplified ratio is 3:2.

40 [Challenging] • Target: Unit rates • Cross-Lesson: 2.01 Quadratics & Polynomials

A chemist mixes 30 mL of a 20% saline solution with 20 mL of a 40% saline solution. What is the saline concentration percentage of the resulting mixture?

a. **28%** ✓

b. 25%

c. 30%

d. 36%

Calculate the total amount of pure saline solute: - From Solution 1: $30 \times 0.2 = 6.0$ mL - From Solution 2: $20 \times 0.4 = 8.0$ mL Total solute = 14.0 mL. Total volume = $30 + 20 = 50$ mL. Concentration percentage: x .

41 [Challenging] • Target: Percentage change and applications • Cross-Lesson: 3.01 Functions & Graphs

According to the data published by “Worldometer”, the total cases of people infected by Covid-19 in Egypt on July 24, 2020, was 91,072, while in August 3, 2020 it went up to 94,640 cases. If we consider the graph of the change in number of total cases over the change in days is linear, what will be the total cases reported in August 9, 2020? (Note that there are 31 days in July).

a. **96,781 cases** ✓

b. 97,138 cases

c. 98,208 cases

d. 101,776 cases

Rate $= (94{,}640 - 91{,}072)/10 = 356.8$ cases/day (10 days from Jul 24 to Aug 3). Aug 9 is 6 more days: $94{,}640 + 6(356.8) \approx 96{,}781$.

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42 [Challenging] • Target: Weighted averages • Cross-Lesson: 5.01 Statistics & Probability

A survey of 300 parents asked which news source they trust most: TV News 102, Newspapers 45, Social Media 87, Radio 66. What percentage of parents trust social media most?

a. 15%

b. 22%

c. **29%** ✓

d. 34%

Applying Ratios, Rates & Data principles, $87/300 \times 100 = 29\%$. Therefore, the verified correct option is 29%.

43 [Challenging] • Target: Scale and similar figures • Cross-Lesson: 6.01 Geometry

On a math test of 30 questions, Mona got 75% of the 12 geometry questions correct, 60% of the 10 algebra questions correct, and 25% of the 8 trigonometry questions wrong. What percentage of all the questions did Mona get correct?

a. 56.6%

b. 160%

c. **70%** ✓

d. 210%

Correct: geometry $0.75 \times 12 = 9$, algebra $0.60 \times 10 = 6$, trig $0.75 \times 8 = 6$. Total $= 21/30 = 70\%$.

44 [Challenging] • Target: Interpreting data tables • Cross-Lesson: 7.01 Trigonometry & Additional Topics

The table shows a city's population for two years. Which statement correctly describes the percent change from 2018 to 2020?

a. **Decreased by 15%** ✓

b. Increased by 15%

c. Decreased by 10%

d. Increased by 10%

$\frac{34000 - 40000}{40000} \times 100 = -15\%$: decreased by 15%.

45 [Challenging] • Target: Bar graphs and data displays • Cross-Lesson: 8.01 Exponentials & Logarithms

A scale model of a building is built at a scale of $1:200$. If the model is 65 cm tall, what is the actual height of the building in meters?

a. 130 ✓

b. 13

c. 1300

d. 65

Solving through rates and algebraic proportions yields 130.

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